DATA ANALYSIS REPORT

Walmart Sales Analysis

Exploratory analysis of weekly store sales, seasonality and outliers

PERIOD	5 Feb 2010 – 26 Oct 2012
DATASET	6,435 weekly records | 45 stores | 8 columns
TOOLS	Python | pandas | seaborn | matplotlib

1. Dataset Overview & Objective

RECORDS
6,435
weekly store observations

STORES
45
store IDs 1 to 45

COLUMNS
8
before and after cleaning

PERIOD
Feb 2010
to Oct 2012 (weekly)

AVG WEEKLY SALES
1.05 M
per store-week (1,046,965)

MAX WEEKLY SALES
3.82 M
3,818,686 in one store-week

Item	Detail
Source file	walmart.csv (Walmart sales data)
Columns	Store, Date, Weekly_Sales, Holiday_Flag, Temperature, Fuel_Price, CPI, Unemployment
Objective	Clean the Walmart sales data, perform EDA, identify sales trends and customer patterns, find outliers and investigate them, and present findings with charts and recommendations.

2. Data Cleaning Summary

Before vs after comparison

Metric	Before	After
Rows	6,435	6,435
Columns	8	8
Missing values	0	0
Duplicate rows	0	0
Duplicate store + date pairs	0	0
date type	object (text)	datetime64
Column names	Mixed case (e.g. Weekly_Sales)	snake_case (e.g. weekly_sales)
weekly_sales outliers	34	34 (kept)
temperature outliers	3	3 (kept)
unemployment outliers	481	481 (kept)

Cleaning steps and decisions

1. **Missing values and duplicates.** Every column was checked: no missing values, no duplicate rows and no duplicate store and date pairs, so no rows were removed.

2. **Column names.** All names were changed to lowercase snake_case (e.g. Weekly_Sales became weekly_sales) for consistency.
3. **Data types.** The date column was text and was converted to a real date (day-month-year, dayfirst=True). All other columns already had correct types; holiday_flag stays 0/1 (0 = normal week, 1 = holiday week).
4. **Outliers (IQR method).** See Section 3. All outliers were kept because they are real values and the holiday sales spikes are something a model should learn from.
5. **Output.** After cleaning, summary statistics showed no impossible values, and the data was saved as walmart_cleaned.csv.

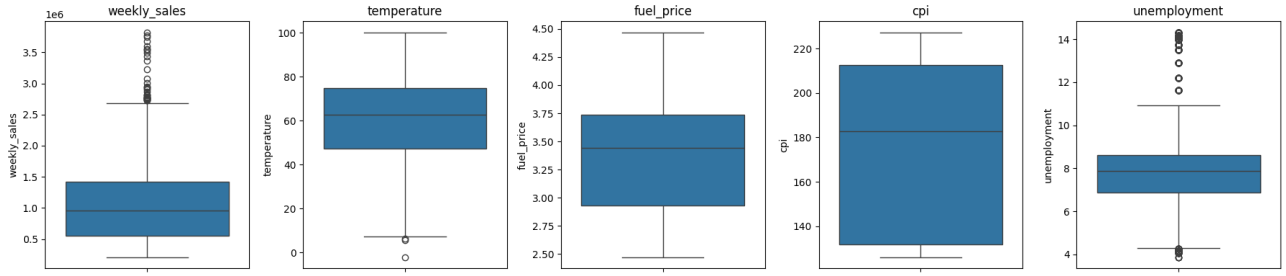
3. Outlier Analysis

Outliers were detected with the IQR method (values below Q1 − 1.5×IQR or above Q3 + 1.5×IQR).

Column	Outliers	Lower bound	Upper bound
weekly_sales	34	-746,862.73	2,720,371.49
temperature	3	6.24	116.16
fuel_price	0	1.73	4.94
cpi	0	10.22	334.26
unemployment	481	4.29	11.22

What the investigation found

- **High weekly sales (34 weeks):** all fall in December (25) or November (9), the holiday shopping season. Only 9 of the 34 are flagged as holiday weeks; 25 are not.
- **High unemployment (378 weeks):** all from stores 12, 28 and 38, so it is a regional feature, not an error.
- **Low unemployment (103 weeks):** all from stores 4, 23 and 40.
- **Decision:** keep all outliers. They are real regional values and genuine seasonal spikes.

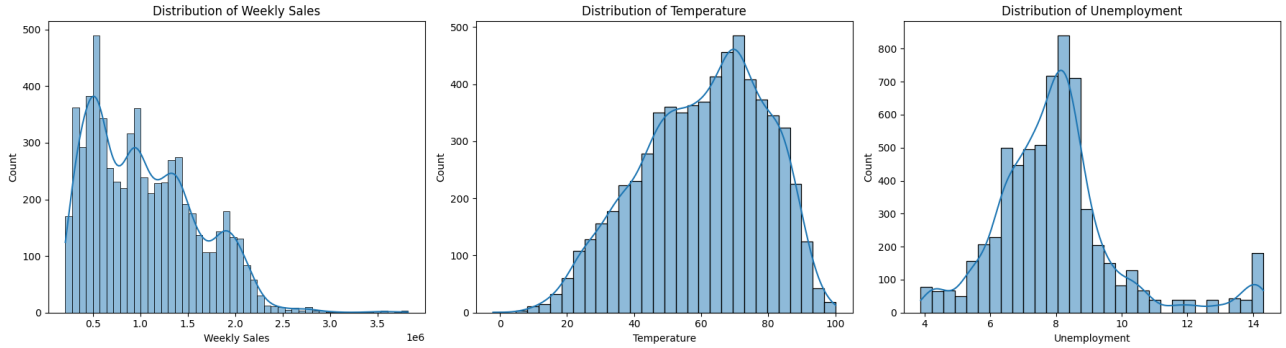


Box plots of the numeric columns. Weekly sales shows a long upper tail of holiday-season weeks, while unemployment shows the separate high and low clusters from a few stores.

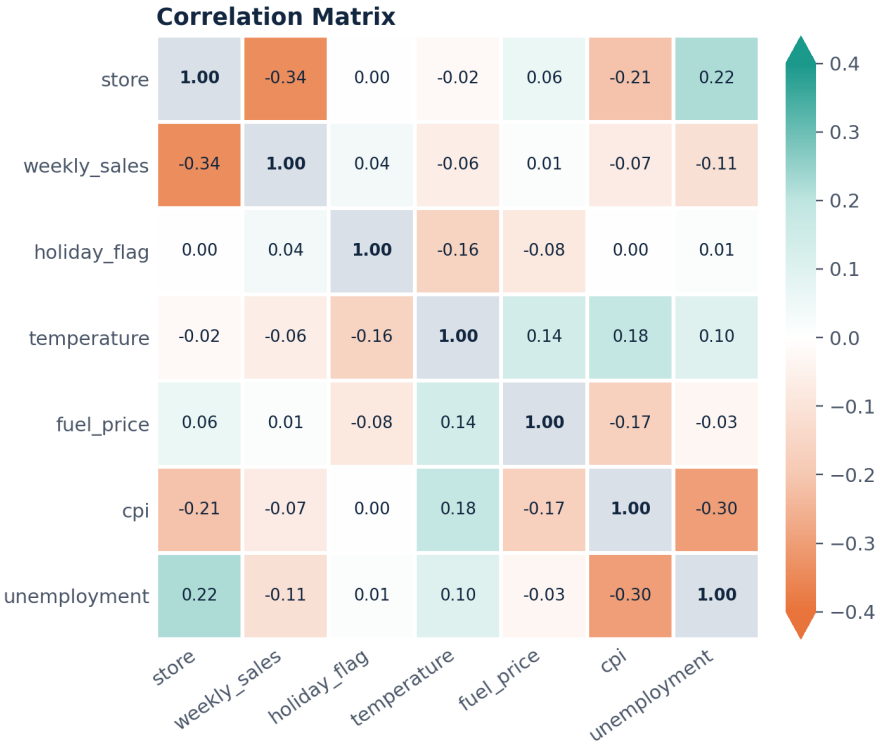
Sample of the highest-sales outliers

Store	Date	Weekly sales	Holiday flag	Unemployment
2	2010-12-24	3,436,007.68	0	8.163
2	2011-12-23	3,224,369.80	0	7.441
4	2010-11-26	2,789,469.45	1	7.127
4	2010-12-17	2,740,057.14	0	7.127
4	2010-12-24	3,526,713.39	0	7.127

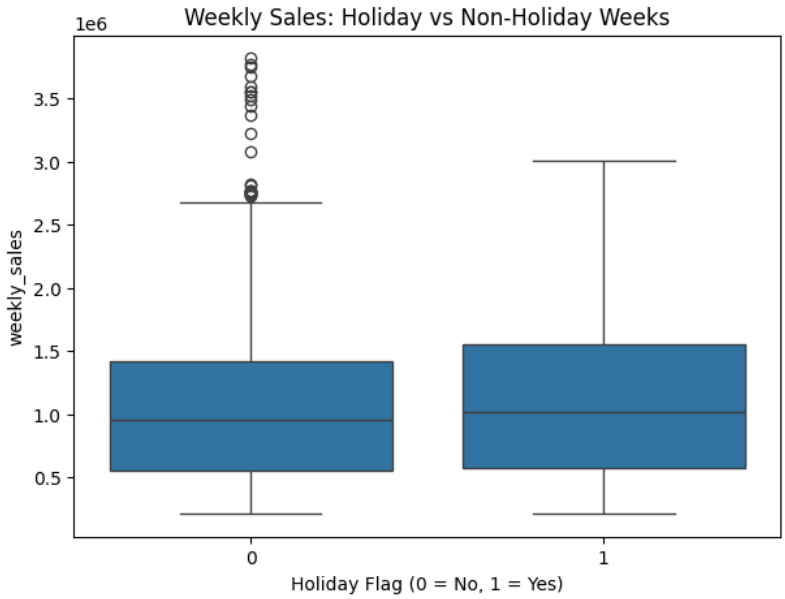
4. EDA and Visualizations



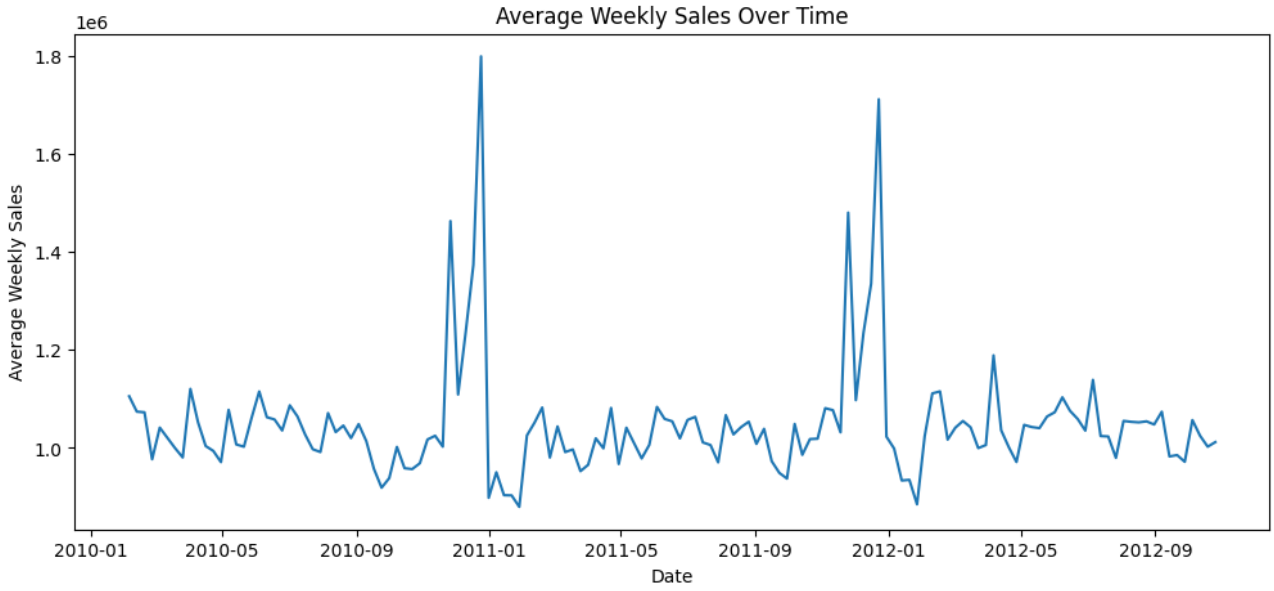
Distributions of weekly sales, temperature and unemployment. Weekly sales are right-skewed: most weeks fall between about 0.3M and 2.0M with a long thin tail up to about 3.8M, and several peaks (around 0.5M, 1.0M, 1.4M and 1.9M), probably because stores differ a lot in size. Temperature is roughly bell-shaped with most weeks between 40°F and 85°F, peaking near 70°F. Unemployment is centered around 8% (mostly 6% to 9%) with a small separate cluster near 14% from a few high-unemployment stores.



Correlation matrix. All correlations with weekly sales are weak: unemployment (-0.11), CPI (-0.07), temperature (-0.06), holiday flag (0.04) and fuel price (0.01), so no single economic factor explains sales on its own. The store column shows -0.34, but store is only an ID number, so that value just tells us stores differ a lot. CPI and unemployment are moderately negatively correlated (-0.30).



Weekly sales: holiday vs non-holiday weeks. Holiday weeks have a slightly higher median and box, but the difference is small. Most of the highest sales (above 3M) are in the non-holiday group, likely because the pre-Christmas week is not flagged as a holiday in this dataset, so the flag misses the biggest sales week.



Average weekly sales over time. Sales stay around 1.0M for most of the year, with clear seasonal spikes every year. One peak is around Thanksgiving (late November) and the largest is just before Christmas (late December), reaching about 1.8M. Sales drop to their lowest right after, in January. There are also small bumps around April.

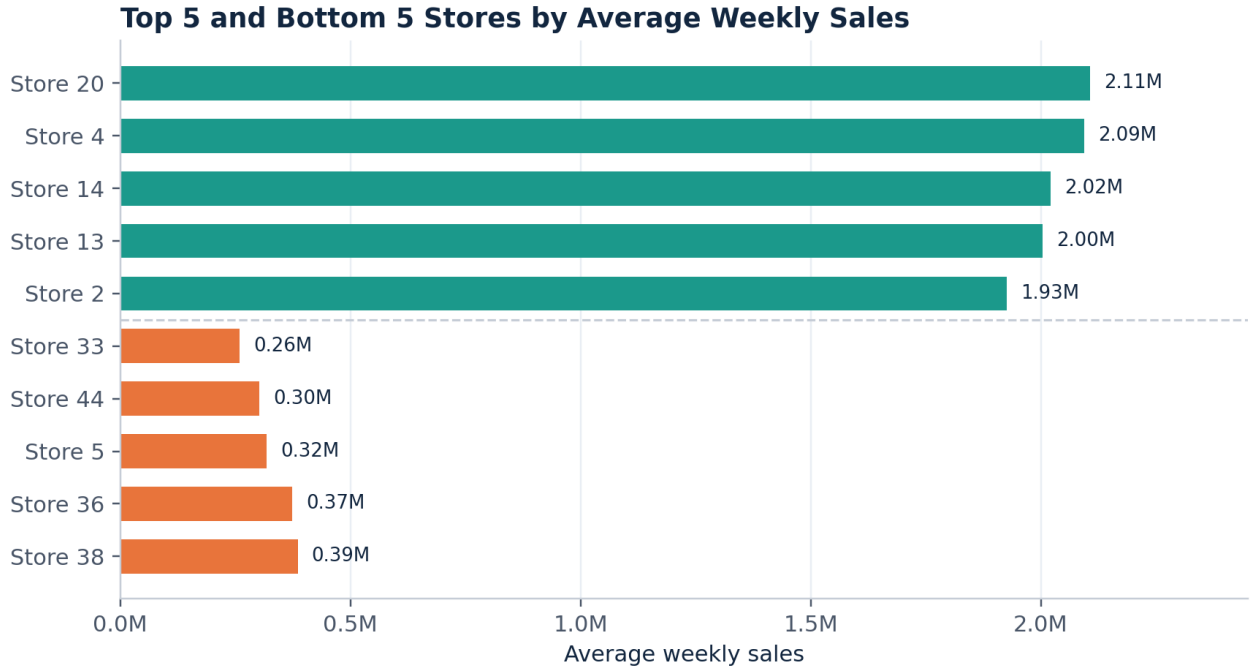


Unemployment vs weekly sales. Higher unemployment goes with lower sales. Above about 10% unemployment almost no weeks go over 2.0M, while the highest sales happen at 6% to 9%. The vertical stripes appear because each store has almost the same unemployment value every week. The relationship is weak, so unemployment is not the only factor.

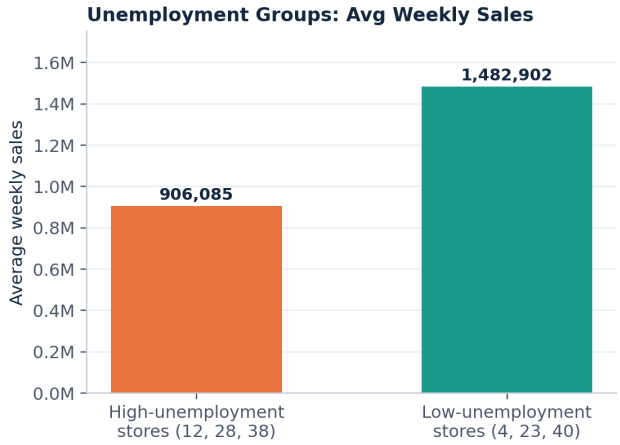
5. Store-Level Analysis



Top 10 stores by average weekly sales. Store 20 ranks first, followed by stores 4, 14, 13 and 2.



Best and weakest stores. Store 20 is the best at about 2.1M average weekly sales. The weakest are stores 33, 44, 5, 36 and 38 (about 260K to 386K). The top store sells about 8 times more than the bottom one.



High vs low unemployment areas. Stores in high-unemployment areas (12, 28, 38) averaged **906,085** per week, while stores in low-unemployment areas (4, 23, 40) averaged **1,482,902**, about **39% lower**. This supports the idea that unemployment is linked to lower sales, but each group has only 3 stores and store size may also play a part.

6. Key Insights & Recommendations

1

Sales are strongly seasonal.

Weekly sales stay around 1.0M for most of the year but spike around Thanksgiving (late November) and peak just before Christmas (late December), reaching about 1.8M. The holiday flag in the dataset does not capture the pre-Christmas week, which is the biggest sales week.

Recommendation: Plan inventory and staffing ahead of late November and late December every year.

2

The store matters more than any economic factor.

Store 20 averages about 2.1M per week, while store 33 averages about 260K, roughly 8 times less. Temperature, fuel price and CPI show almost no correlation with sales (all below 0.1 in absolute value).

Recommendation: Study what top stores like 20 and 4 do differently and apply it to weaker stores.

3

High unemployment is linked to lower sales.

Stores in high-unemployment areas (12, 28, 38) averaged 906,085 per week versus 1,482,902 in low-unemployment areas (4, 23, 40), about 39% lower. The comparison uses only 3 stores per group and store size may also play a part, so it shows a link, not proof of cause.

Recommendation: Use targeted promotions and lower-priced products in high-unemployment areas.

7. Conclusion

Store and season explain most of the variation in Walmart's weekly sales, while weather and fuel price matter very little. Unemployment has a weak negative link with sales, but it is mixed up with store differences.

Next step: a prediction model should use store and date-based features (month, week of year, holiday weeks) rather than relying only on economic variables.